

CITS 1501 Introduction to Programming with Python



Released: Week 6 · **Due:** end of Week 11 · **Weight:** 30% of the unit · **Teams:** 2 students

1. Project Overview

In this project, you will work in a group to **design, build, test, and deploy an interactive app for a specific user or purpose**. You will act as a small development team responsible for turning a project brief into a working application.

Your task is to create an app that helps users **explore information, complete a task, answer a question, or make a decision**. The app should allow users to interact with information and see the results of their actions.

You will:

- choose a project brief;
- identify the intended users and their needs;
- find and prepare suitable data;
- design and implement an interactive application;
- analyse and visualise the data;
- test the application;
- deploy the application;
- document your work; and
- demonstrate and explain your application.

All tools are expected as part of the project, but **you must understand and be able to explain all work you submit**.

2. Project Scenario: Culture, Country and Language

You are working as a small development team creating a digital application for a community, educational organisation, researcher, visitor, or other appropriate user.

Your application should help people learn about or engage with **Aboriginal and Torres Strait Islander culture, history, language, or connection to Country**.

Western Australia has a rich and diverse Aboriginal cultural and linguistic history represented in public collections, language resources, archives, maps, and educational material. These resources can support applications about language, place, history, culture, and related topics.

CITS 1501 Introduction to Programming with Python



Design Briefs

Choose **one** of the following briefs, or propose your own suitable idea:

Design brief	Example application
Language Learning	An interactive tool that helps users learn or explore a published wordlist.
Seasonal Knowledge	An app explaining Indigenous seasonal calendars and associated environmental signs.
Collection Explorer	A browser for publicly available artworks, photographs, recordings, or archival material.
History Explorer	An interactive timeline exploring events, places, or people.
Language and Country Explorer	An application exploring relationships between language, places, and Country.
Population Explorer	An application exploring and comparing published population or regional data.
Your Own Idea	A different application that fits the subject and project requirements.

Your team should decide **who the app is for, what problem it addresses, what users should be able to do, and what information the app needs to provide.**

3. Your data

Your app must use a dataset with **at least 200 records**. The data must be loaded from a file or API rather than typed directly into your code.

Before using the data, you must:

- Check that the dataset contains at least 200 records.
- Check that the data supports your project idea.
- Identify the original source.
- Check copyright and licensing conditions.
- Clean and prepare the data where necessary.
- Explain how the data is used in your application.

CITS 1501 Introduction to Programming with Python



Cultural and Data Use Requirements

When using Aboriginal and Torres Strait Islander material:

- Use material published by a recognised organisation, institution, community organisation, or language centre.
- Clearly identify the source and its usage conditions.
- Do not use AI to generate cultural, language, or historical content.
- Check copyright, licensing, and access restrictions.
- Do not use restricted cultural knowledge.
- Do not use material if the source or permission is unclear.
- Follow relevant requirements when using images, names, or voices of deceased people.
- Ask your facilitator or Unit Coordinator if you are unsure whether material is appropriate.

Table 1. Suggested Data Sources

Source	Website	What it provides
AIATSIS	aiatsis.gov.au	Aboriginal and Torres Strait Islander languages and cultural collections, including AUSTLANG.
South West Aboriginal Land and Sea Council	noongar.org.au	Noongar culture, language, Country and educational resources.
Noongar Boodjar Language Cultural Aboriginal Corporation	noongarboodjar.com.au	Noongar language resources and protocols.
Bureau of Meteorology	bom.gov.au	Indigenous seasonal calendars and weather knowledge.
Australian Bureau of Statistics	abs.gov.au	Census and population data.
State Library of WA	slwa.wa.gov.au	Digitised collections and archives.
Trove	trove.nla.gov.au	Digitised collections, newspapers and photographs.

CITS 1501 Introduction to Programming with Python



Source	Website	What it provides
Collections WA	collections.wa.net.au	Western Australian collections and cultural resources.
Whadjuk Aboriginal Corporation	whadjuknoongar.org.au	Information about Whadjuk Country and the Perth area.
data.gov.au	data.gov.au	Australian government datasets.
data.wa.gov.au	data.wa.gov.au	Western Australian government datasets.
Kaggle	kaggle.com/datasets	Public datasets; check the original source and licence.

Note: These sources are starting points rather than guaranteed suitable datasets. Teams must check that their chosen dataset meets the project requirements and is appropriate and legal to use.

4. Technical Expectations

Your application should demonstrate your ability to apply the programming and software development skills taught in CITS1501.

The application should:

- contain clear, readable, and structured code;
- implement the main functionality using your own code;
- have an appropriate software structure;
- handle invalid input and reasonable edge cases;
- use GitHub throughout development;
- include contributions from every team member; and
- include a README.md explaining how to install, run, use, and test the application.

5. Analysis and Visualisation

Your application should do more than simply display raw data. It should analyse the data to produce meaningful information and communicate the results clearly to users. Depending on your project, this may include counts, averages, rates, comparisons, rankings, changes over time, trends, or other appropriate calculations. The results should be presented using suitable visualisations, such as charts, maps, timelines, tables, or summary figures. Your visualisations

CITS 1501 Introduction to Programming with Python



should help users understand patterns or insights that are not immediately obvious from the raw data. Assessment will focus on the correctness and relevance of your analysis, the appropriateness of your visualisations, the accuracy of your interpretation, and how clearly you communicate the results to users.

6. User Interface and Experience

The final application must provide a clear and user-friendly interface.

It must include:

- at least three distinct screens or views;
- clear navigation;
- meaningful user interaction;
- appropriate feedback when users interact with the app;
- a consistent and suitable visual design; and
- an interface appropriate for the intended users.

Users should be able to perform the main task without unnecessary difficulty.

7. Algorithms and Problem Solving

Your application must include meaningful computational work. The main algorithm should do more than simply call a library function that performs the entire task.

Your team should be able to explain:

- what the algorithm does;
- why it is appropriate for the problem;
- how it processes the data;
- how it handles important edge cases; and
- why you made your main design decisions.

8. Testing and Reliability

Your team must provide at least eight automated tests.

Tests should cover:

- the main functionality;
- the algorithmic core;
- invalid input;
- important edge cases; and

CITS 1501 Introduction to Programming with Python



- boundary conditions where appropriate.

The application should not crash during reasonable use.

Your team should also test the main user workflows and fix problems identified during testing.

9. Architecture

Your project must include an architecture diagram showing:

- the main components of the application;
- how the components interact; and
- how data moves through the system.
- The diagram should accurately represent the application you actually built.

10. Security and Privacy

Your team should identify and address security and privacy risks relevant to your application.

You should:

- never commit passwords or API keys to GitHub;
- avoid collecting unnecessary personal information;
- validate user input where appropriate;
- handle stored information responsibly; and
- consider relevant security and privacy risks.

11. Deployment

The final application must be available for someone outside your development environment to use.

For example:

- a web application should normally be hosted online; or
- a mobile application should have an installable version that can be demonstrated.

The deployment should include everything required for the application to operate.

12. Use of AI

AI tools are permitted and expected in this project.

You must:

- understand the code you submit;

CITS 1501 Introduction to Programming with Python



- check AI-generated code before using it;
- test AI-generated solutions;
- modify or reject AI output when necessary; and
- maintain an AI-LOG.md documenting significant AI use.

13. Report

Submit a technical report of no more than 3,000 words as a PDF.

The report should cover:

1. Team members and GitHub link.
2. What you built and who it is for.
3. Architecture and data flow.
4. Algorithms and technical decisions.
5. Data source and preparation.
6. Analysis and visualisation.
7. Interface and user experience.
8. Testing and limitations.
9. Security and privacy.
10. Deployment.
11. Team contribution.

14. Demo and Q&A

In Week 12, your team will demonstrate the application and answer questions.

The demonstration should show:

- the working application;
- a realistic user task;
- important features and interactions;
- an invalid input or failure case; and
- your automated tests.

During the Q&A, students may be asked to explain any part of the project, including code written by another team member or AI-assisted code.

CITS 1501 Introduction to Programming with Python



15. Submission

What	When	How	Who
Report + Github link (PDF)	End of Week 11	LMS	One per team
Demo and Q&A	Week 12	In person	Whole team

16. Assessment Rubric — 100 Marks

Assessment Area	Marks	What is assessed
Technical Development and Code Quality	8	Clear, readable, structured, maintainable code and correct implementation.
Data	7	Appropriate dataset, 200+ records, preparation, processing, source, licensing, and responsible use.
Algorithms and Problem Solving	8	Meaningful algorithmic work, correct implementation, appropriate problem-solving, and justification of decisions.
Analysis and Visualisation	7	Meaningful analysis, correct calculations, appropriate visualisations, and clear communication of results.
Architecture	7	Appropriate system structure, components, data flow, and accurate architecture diagram.
User Interface and Experience	7	At least 3 screens/views, clear navigation, meaningful interaction, and appropriate user experience.
Testing and Reliability	7	At least 8 automated tests, edge cases, error handling, and reliable application behaviour.
Security and Privacy	5	Appropriate handling of security, privacy, user data, passwords, API keys, and other risks.
Deployment	5	Working deployment that can be used outside the development environment.
Use of AI	4	Appropriate AI use, AI log, verification of AI output, and understanding of submitted code.
Technical Report	10	Clear explanation of the application, architecture, algorithms, data, analysis, design, testing, security, deployment, and teamwork.

CITS 1501 Introduction to Programming with Python



Assessment Area	Marks	What is assessed
Demo and Q&A	10	Working demonstration, realistic task, failure/edge case, tests, and ability to explain technical decisions.
Individual Contribution and Understanding	15	Git contributions, meaningful work, contribution statement, teamwork, and understanding of the whole project.
Total	100	

Individual Contribution — 15 Marks

Individual assessment	Marks
GitHub contribution and meaningful work	5
Individual contribution statement	3
Understanding of the whole project	5
Contribution to team and professional collaboration	2
Total	15

Grade Bands

Grade	Mark	Overall Standard
High Distinction (HD)	80–100%	Excellent work that is complete, technically strong, well justified, and demonstrates a high level of independence and attention to detail.
Distinction (D)	70–79%	Very good work that meets the requirements well, with sound technical and design decisions and only minor weaknesses.
Credit (C)	60–69%	Good work that meets the main requirements and demonstrates competent implementation, but has some limitations or areas that could be improved.
Pass (P)	50–59%	Satisfactory work that meets the basic requirements but has significant limitations in implementation, reliability, design, explanation, or completeness.
Fail (F)	0–49%	Work that does not adequately meet the project requirements, contains major technical or conceptual problems, or provides insufficient evidence of the required work.